DELIVERABLE D2.4

Data Management Plan

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SEMDR

Smart Energy Management and Demand Response

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Abstract

This report describes the data management practices adopted throughout the project, covering data collection, processing, storage, security, GDPR compliance, and the procedural framework for preparing data for open access publication.

The SEMDR project deploys an IoT-based energy monitoring and demand response platform at a hotel demonstration site in Cyprus. The system collects real-time energy consumption data from the main hotel supply and its principal energy-intensive assets, complemented by environmental sensing, weather observations, and Cyprus day-ahead electricity market prices. All data is transmitted securely from the edge to a cloud platform where it is validated, stored, and made available to the project's analytics layer. Detailed specifications of the system architecture and data collection infrastructure are documented in the companion technical deliverables D4.1 (Data Collection and Analysis Report) and D5.1 (System Design and Development Report).

This Data Management Plan documents the complete data lifecycle at a policy and procedural level: the data categories collected and generated; the FAIR principles (Findable, Accessible, Interoperable, Reusable) that govern how data is described, stored, and shared; the security and retention principles applied throughout the system; the GDPR compliance framework confirming that no personal data is collected or processed; and the procedural framework for anonymising and preparing datasets for open publication. The specific open science outputs, dataset inventories, and release timelines are documented separately in D2.3 (Open Science Activities Report), while this DMP describes the methods and procedures that support those commitments.

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1 Introduction

This section introduces the purpose and scope of the SEMDR Data Management Plan, explains its relationship to the project's Open Science commitments, and establishes its position within the broader deliverable portfolio.

1.1 Purpose and Scope

The purpose of this document is to provide a single, authoritative reference for how all data within the SEMDR project is collected, validated, stored, protected, and - where appropriate - prepared for open publication. It is intended for use by the consortium partners, the Research and Innovation Foundation, and any external reviewer who needs to understand the project's data governance arrangements at a policy and procedural level.

The scope covers the entire data lifecycle, from the capture of sensor readings at the demonstration site through to the long-term archival and potential open publication of research datasets. Specifically, the plan addresses the categories of data collected (Section 2), the FAIR principles governing data description and sharing (Section 3), the infrastructure and security principles applied to data in transit and at rest (Section 4), the GDPR compliance framework (Section 5), the procedures for anonymising and sharing data (Section 6), the long-term archiving strategy and the roles and responsibilities of each consortium partner (Section 7).

This DMP is a living document, updated throughout the project to reflect evolving data management practices. It is complemented by several companion deliverables, each covering a distinct aspect of the project's data and dissemination activities. D2.3 (Open Science Activities Report) documents the specific open science outputs, including dataset inventories, release timelines, and embargo periods. This DMP describes the procedures, while D2.3 describes the specific outputs. D2.2 (Commercialisation Plan) defines the IPR framework governing the boundary between proprietary and open content. D4.1 (Data Collection and Analysis Report) and D5.1 (System Design and Development Report) contain the complete technical specifications of the data collection infrastructure, system architecture, and data pipeline, which are referenced throughout this DMP rather than duplicated here.

1.2 Relationship to Open Science

The SEMDR project is committed to the principles of Open Science as promoted by the Research and Innovation Foundation and the European Commission. This commitment is reflected in the project's approach to data sharing, which balances the need for open access to research outputs with the protection of commercially sensitive information and the intellectual property rights of the consortium partners.

The project follows an "as open as possible, as closed as necessary" approach, ensuring that data which could enable further research and innovation is shared openly, while data that contains proprietary algorithms or commercially sensitive operational parameters is appropriately protected. The specific open science outputs including dataset inventories,

release timelines, embargo periods, and repository commitments, are documented in D2.3. This DMP describes the procedural and technical framework for how data is managed, anonymised, and prepared for sharing; D2.3 documents what specific outputs are shared, when, and where. This separation ensures that if the open science plan evolves during or after the project, only D2.3 requires updating rather than this Data Management Plan.

2 Data Summary

This section provides a high-level overview of the categories of data collected, processed, and generated within the SEMDR project, along with the formats and standards applied.

2.1 Overview of Data Collected

The SEMDR project generates and processes several categories of data throughout its lifecycle. The primary data source is the real-time energy monitoring system deployed at the hotel demonstration site, complemented by external weather data, Cyprus day-ahead electricity market prices, and analytical outputs generated by the project's optimisation algorithms.

Table 1 provides a summary of the principal data categories. The specific sensor hardware, acquisition protocols, and database schemas used to capture and store this data are documented in D4.1 and D5.1.

Data Category	Source Type	Description
Energy consumption	On-site metering	Electrical measurements from the hotel's main supply and its principal energy-consuming assets: active power (kW), energy (kWh), and related electrical parameters.
Environmental sensing	On-site sensors	Ambient temperature and humidity in relevant indoor spaces; water quality parameters (pH, ORP, temperature) for the pool system.
Building management data	BMS integration	Read and write access to the hotel's Building Management System via the industry-standard oBIX protocol, providing operational context and control capability for the HVAC system.
Weather observations	External API	Outdoor weather conditions used as input to predictive models. Subject to third-party redistribution restrictions (see Section 6.4).
Electricity market prices	TSOC public data	Cyprus day-ahead market clearing prices (48 half-hourly slots per day, EUR/kWh). Publicly available market data.
Analytical outputs	Generated by platform	Forecasts, recommended operating schedules, and actuation records produced by the project's optimisation algorithms.
System diagnostics	Cloud platform	Infrastructure monitoring, device heartbeats, data quality flags, and alert history.

Table 1 - SEMDR Data Categories

2.2 Data Formats, Standards and Volume

Sensor telemetry is transmitted in structured JSON format over encrypted connections and stored in a relational database with standardised schemas. For data export and sharing purposes, datasets are converted to open, non-proprietary formats (CSV for tabular data, JSON for structured data, IDF for building simulation models) with comprehensive metadata and accompanying documentation.

Time-series data follows the ISO 8601 standard for all timestamps, with all times recorded in UTC to avoid ambiguity. Energy measurements are recorded in kilowatt-hours (kWh) for consumption and kilowatts (kW) for instantaneous power, following SI unit conventions. Temperature readings

are recorded in degrees Celsius. All numerical values use decimal point notation with appropriate precision for each measurement type.

The SEMDR monitoring infrastructure has been operational since M5 (August 2025), generating approximately one year of continuous time-series data across all active streams by the end of the project period. Data is stored in the project's production database for active use, with raw sensor payloads also archived to cloud object storage as a recovery and audit resource. Volume, growth rate, and storage architecture are documented in D5.1.

3 FAIR Data Principles

The SEMDR project adheres to the FAIR data principles (Findable, Accessible, Interoperable, Reusable) [1] as recommended by the European Commission and the Research and Innovation Foundation. These principles guide every aspect of data management, from how metadata is recorded at the point of collection through to how datasets are prepared for long-term preservation and open publication. The implementation of these principles for the specific open science outputs produced by the project is documented in D2.3 (Open Science Activities Report). Table 2 summarises the general framework that applies across all project data.

FAIR Principle	Implementation
Findable	All datasets published through open repositories are assigned persistent identifiers (DOIs). Comprehensive metadata records are created using the Dublin Core standard, including descriptions, keywords, and provenance information. Datasets are registered in Zenodo and indexed by OpenAIRE.
Accessible	Published datasets are accessible via standard HTTPS protocols from Zenodo and, where relevant, from a public GitHub repository. Non-sensitive datasets are freely available under open licences. Access conditions are clearly stated in each dataset record.
Interoperable	Data is stored and shared in widely-used, open formats (CSV, JSON, IDF) that are machine-readable and tool-agnostic. Metadata follows established schemas (Dublin Core, DataCite). Time-series data uses ISO 8601 timestamps and SI units for all measurements.
Reusable	All shared datasets include comprehensive documentation describing collection methods, data structure, and any processing steps applied. Data is released under Creative Commons CC BY 4.0 licences; software is released under MIT. Quality assurance procedures and data processing steps are fully documented.

Table 2 - FAIR Principles Implementation

3.1 Metadata Standards

All datasets published through open repositories are accompanied by comprehensive metadata records following the Dublin Core Metadata Standard. Metadata includes, at a minimum: title, creator(s), description, subject keywords, date of creation, data format, rights and licence information, and a link to the parent project (SEMDR, COM-CONCEPT-ENERGY/0624/0160). Additional domain-specific metadata includes sensor and measurement specifications, measurement units, sampling rates, and any relevant contextual information that supports interpretation and reuse.

For time-series energy data, accompanying metadata files include the asset category, measurement units, data collection period, sampling frequency, and any known data gaps or quality issues. Each sensor reading carries a quality indicator populated by automated outlier and completeness checks, ensuring that downstream users of any shared dataset can identify records that were flagged during collection.

4 Data Infrastructure and Security

This section describes the principles governing the project's data collection infrastructure, cloud-based storage, security architecture, and retention policy. The full technical specifications of the hardware, cloud services, database schema, network topology, and security implementation are documented in the companion deliverables D4.1 (Data Collection and Analysis Report) and D5.1 (System Design and Development Report). This section summarises the relevant principles at the level appropriate for a public-facing Data Management Plan.

4.1 Data Infrastructure

The data collection infrastructure was installed at the hotel demonstration site by Restrol (PA2), the consortium's hardware integration partner. The installation comprises IoT monitoring sensors and data acquisition modules connected to the hotel's main supply and its principal energy-consuming assets, along with environmental and water quality sensors. All on-site devices communicate with the project's cloud platform via a dedicated wireless network that is fully isolated from the hotel's guest-facing IT infrastructure. The specific hardware models, network configuration, and integration interfaces are documented in D4.1 and D5.1.

Sensor measurements are captured at the edge, transmitted securely over encrypted connections to the project's cloud platform, validated against schema and plausibility constraints, and stored in a managed relational database hosted within the EU region. Edge devices maintain local buffering to ensure no data is lost during temporary network interruptions, and every measurement carries a unique identifier that allows duplicate detection at the database layer. Raw payloads are also archived to cloud object storage as a secondary recovery and audit resource.

The analytics layer runs on a secure cloud application platform and executes the project's optimisation algorithms on a scheduled basis, writing recommended operating schedules back to the database. A policy engine checks each recommendation against hard safety guardrails before any command is issued to the hotel's equipment. The detailed pipeline architecture, cloud services, database schema, and scheduling implementation are documented in D5.1.

Data quality is assured through a combination of automated and manual processes. The automated validation pipeline applies four categories of check to incoming data at the point of ingestion:

- **Range validation:** each sensor type has defined acceptable ranges; readings outside these ranges are flagged as potential sensor faults.
- **Outlier detection:** statistical outlier tests identify anomalous readings that, while within the range limits, deviate significantly from recent patterns; these are flagged in the data record.
- **Completeness monitoring:** a dedicated monitoring function continuously scans each data stream for gaps; any significant gap triggers an alert for engineer investigation.

- **Periodic reviews:** monthly data quality reviews are conducted by the technical team, examining overall data completeness rates, sensor availability statistics, and recurring issues. These reviews are documented and shared with the consortium during progress meetings.

The specific thresholds, algorithms, and review procedures applied within each category are documented in D5.1.

4.2 Security Architecture

Protecting project data and ensuring that only authorised devices and users can interact with the system is essential for both operational safety and regulatory compliance. The security architecture addresses this through multiple layers of protection, summarised in Table 3 at a principles level. The complete security implementation, including specific protocols, authentication mechanisms, and credential management approach, is documented in D5.1.

Security Layer	Principle
Encryption at Rest	All stored data is encrypted using industry-standard symmetric encryption with centrally managed keys and regular key rotation.
Encryption in Transit	All communication between edge devices and the cloud platform, and between cloud services, uses industry-standard transport layer encryption with mutual authentication where appropriate.
Device and User Authentication	Each edge device has a unique cryptographic identity, and access to cloud services is governed by role-based access control with multi-factor authentication for administrative roles.
Network Isolation	On-site monitoring devices operate on a dedicated network segment, isolated from the hotel's guest-facing IT infrastructure. Only the specific protocols required for system operation are permitted outbound, and no inbound connections from the public internet are accepted.
Audit and Monitoring	All significant actions on the platform are logged with actor identity and timestamp. Continuous monitoring of infrastructure and data stream health is in place, with automated alerts configured for exceptional conditions.
Backup and Recovery	Automated daily backups of the production database with 30-day retention. A secondary archive of raw sensor payloads provides an independent recovery path. Recovery procedures are tested periodically.

Table 3 - Security Architecture Principles

Project data will be retained for a minimum of five years following the end of the project (until at least March 2031), in accordance with the requirements of the Research and Innovation Foundation and EU data retention guidelines for funded research projects. The retention strategy operates in two phases: active database retention for approximately 24 months post-project, followed by migration to cloud cold storage for the remainder of the five-year commitment. This two-stage approach balances query performance during the active analysis period with cost-effective long-term preservation.

After the mandatory retention period, data will be reviewed for continued scientific or commercial value. Datasets published in open repositories (Zenodo) will remain available indefinitely as per the repository's preservation policy. Commercially sensitive data that is no longer required will be securely deleted using industry-standard data destruction procedures.

5 GDPR Compliance

This section documents the legal basis for data processing within the SEMDR project, the data minimisation measures adopted, and the data processing agreements between consortium partners and the pilot hotel.

The processing of data within the SEMDR project is conducted under the legal basis of legitimate interest (Article 6(1)(f) GDPR [1]) for energy system monitoring and optimisation, and contractual necessity (Article 6(1)(b) GDPR [1]) in relation to the consortium agreement and the hotel pilot agreement.

It is important to note that the SEMDR energy monitoring system does not collect, process, or store any personal data relating to hotel guests. The monitoring equipment is installed on building-level energy systems and does not monitor individual room energy consumption or any guest-identifiable information. The project has confirmed through its data processing register that no Article 9 special category data (as defined in GDPR) - including data relating to health, biometric identity, racial or ethnic origin, political opinions, or religious beliefs - is collected or processed within any SEMDR data stream. All monitored parameters are physical measurements of building systems and environmental conditions with no link to identifiable individuals.

The project adheres to the principle of data minimisation by collecting only the data that is strictly necessary for achieving the project's scientific and technological objectives. In practice this means: only aggregate, asset-level energy consumption is collected, with no sub-metering at the individual room level; only the total hotel occupancy rate (as a percentage) is used for algorithm context, provided as a single daily value with no link to guest booking data, names, or other personal identifiers; and any feedback collected from hotel staff regarding system performance is anonymised at the point of collection.

Formal data processing agreements have been established between the consortium partners, defining the roles and responsibilities of each partner with respect to data handling. IMKA, as the host organisation, serves as the Data Controller under GDPR. GridQ (PA1), which operates the cloud platform and analytics engine, acts as the Data Processor. Restrol (PA2), responsible for hardware installation and maintenance, has limited data access restricted to sensor configuration and diagnostic data.

6 Data Sharing and Open Access

This section describes the procedural framework for preparing and sharing research data from the SEMDR project. It covers the open access commitment, the categories of data eligible for open publication, the anonymisation procedures that prepare data for release, and the restrictions that protect commercially sensitive information. The specific open science outputs, including dataset identifiers, release timelines, and repository commitments, are documented in D2.3 (Open Science Activities Report).

6.1 Open Access Commitment

The SEMDR project is committed to making research data and publications openly accessible, in line with the principles of the Cyprus Open Science Practices framework and the EU Open Science Policy. All peer-reviewed publications resulting from the project are published in open access journals or deposited in open access repositories.

The project's open access commitment is governed by two principles. First, research outputs that could benefit the wider scientific and engineering community, including anonymised datasets, non-proprietary software tools, and methodology documentation, are shared openly through established repositories with persistent identifiers and clear citation guidance. Second, proprietary algorithm implementations, commercially sensitive operational data, and information protected under the consortium's IPR framework (documented in D2.2, Commercialisation Plan) are appropriately restricted.

6.2 Data Categories Eligible for Open Publication

The types of data that may be prepared for open publication fall into three broad categories.

- **Anonymised energy consumption time-series** - Aggregated and anonymised energy consumption data from the hotel demonstration site, suitable for use by researchers working on energy management, demand response, or building performance analysis. Data is published in CSV format with comprehensive metadata headers and accompanying README documentation following the Dublin Core metadata standard.
- **Algorithm performance and methodology documentation** - Non-proprietary descriptions of the demand response scheduling approach, including performance metrics, safety constraint designs, and validation results. Proprietary algorithm parameters, training data, and optimisation strategies remain protected. The boundary between proprietary and open content is defined in D2.2.
- **Software tools and simulation artefacts** - Non-sensitive software components and simulation assets - such as data acquisition scripts, price forecasting tools, and building simulation models built on publicly available reference buildings - may be released under permissive open source licences with documentation and DOI-based archival. Specific outputs in this category are documented in D2.3.

6.3 Anonymisation Procedures

Before any dataset derived from the hotel demonstration site is released for open publication, it undergoes a defined anonymisation procedure designed to preserve scientific value while removing information that could identify the site or expose commercially sensitive operational detail. The procedure includes the following steps:

- **Identifier removal:** the hotel name, GPS coordinates, street address, and any references to the parent organisation or branded equipment are removed from all metadata, filenames, and data content.
- **Temporal shifting:** where appropriate for the research purpose, absolute calendar dates may be replaced with relative day indices (day 1, day 2, etc.) to prevent linkage of operational patterns to specific dates. Temporal structure (time of day, day of week) is preserved where scientifically relevant.
- **Equipment detail abstraction:** specific equipment model numbers, serial numbers, and vendor-specific identifiers are removed or replaced with generic descriptors (for example, "air-cooled chiller, nominal capacity approximately 250 kW" rather than a specific vendor model number).
- **Pattern review:** data is reviewed for patterns that could reveal commercial operational schedules or occupancy-linked behaviour that the hotel considers confidential. Where such patterns exist, aggregation or masking is applied to preserve the statistical relationships relevant to research without exposing the underlying operational detail.
- **Final review and approval:** before publication, each anonymised dataset is reviewed by a consortium representative from IMKA (as Data Controller) and a technical representative from GridQ (as Data Processor) to confirm that the anonymisation is complete and that no residual identifying information remains.

Datasets are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence upon publication, allowing for maximum reusability while ensuring appropriate attribution to the SEMDR project.

6.4 Restrictions and Embargo Periods

Certain categories of data are subject to access restrictions to protect the commercial interests of the consortium partners and the operational privacy of the pilot hotel.

- **Proprietary algorithm parameters:** The specific configuration parameters, training data, and optimisation strategies of the project's demand response algorithms are classified as trade secrets and will not be published. The algorithmic methodology at a conceptual level and the performance results are described in the project's technical reports and open access publications. The boundary between proprietary and open content is defined in D2.2.
- **Detailed site-specific operational data:** Granular operational data that could reveal commercially sensitive information about the hotel's energy costs, operational schedules, or guest occupancy patterns is not shared in raw form. Such data is shared only in aggregated or anonymised form, using the procedures described in Section 6.3.

- **Third-party data with redistribution restrictions:** Data consumed from third-party services that restrict redistribution is excluded from open publication in accordance with the relevant terms of service. Where alternative openly redistributable data sources exist, those are used for open science outputs instead.
- **Embargo periods:** Datasets scheduled for open publication may be subject to embargo periods to allow time for additional processing, anonymisation, peer-reviewed publication, and consortium review before public release. Specific embargo durations for each output are documented in D2.3.

7 Archiving, Roles and Responsibilities

This section describes the long-term preservation strategy for SEMDR project data and the roles and responsibilities of each consortium partner in implementing this Data Management Plan.

7.1 Long-Term Preservation

The long-term preservation of SEMDR project data is ensured through a dual approach: cloud-based archival storage maintained by GridQ (PA1) on the project's cloud infrastructure, and deposition in the Zenodo open access repository operated by CERN.

Within the pilot deployment, raw sensor payloads are archived automatically to cloud object storage, providing an immediate archival and recovery resource. The production database retains active data for the project duration and approximately 24 months post-project, after which data is migrated to cloud cold storage for long-term preservation. Together, these phases fulfil the five-year post-project retention commitment described in Section 4.5. Archived data is stored in open, non-proprietary formats (CSV for tabular data, JSON for structured data, IDF for building simulation models, PDF/A for documentation) to ensure long-term readability and accessibility regardless of future software changes. All archived datasets are accompanied by comprehensive metadata and README files describing the data structure, collection methods, and any processing applied.

Zenodo hosts the open access datasets indefinitely, providing persistent storage and DOI-based referencing. Software source code released under open source licences is preserved through the GitHub-Zenodo integration, which mints a persistent DOI for each tagged release and ensures the contents are archived in CERN's infrastructure regardless of any future changes to the GitHub hosting arrangement.

7.2 Roles and Responsibilities

Clear assignment of data management roles and responsibilities is essential for effective implementation of this Data Management Plan. Table 4 summarises the key responsibilities of each consortium partner.

Partner	Role	Responsibilities
IMKA (HO)	Data Controller	Overall responsibility for GDPR compliance, data governance, and the Data Management Plan. Manages data processing agreements, coordinates open data publications, oversees data archiving and long-term preservation.
GridQ (PA1)	Data Processor	Technical implementation of the data pipeline, cloud storage, and analytics platform. Develops data quality assurance procedures, manages database infrastructure, and implements security controls. Leads the preparation of datasets for open publication.
Restrol (PA2)	Data Collector	Installation and maintenance of IoT monitoring hardware and sensors. Responsible for data collection integrity at the hardware level, sensor calibration, and ensuring reliable data transmission from on-site equipment to the cloud platform.

Table 4 - Data Management Roles and Responsibilities

The Data Management Plan is overseen by the Project Coordinator (Dr. Andrey Sesyuk, IMKA), who is responsible for ensuring all partners adhere to the agreed data management procedures. Any questions or issues relating to data management should be directed to the Project Coordinator in the first instance. Data management practices are reviewed during quarterly consortium meetings, with any necessary updates to this DMP agreed by all partners. Significant changes to data management procedures require formal approval from the project coordinator and are documented in updated versions of this plan.

References

[1] European Commission, "Guidelines on FAIR Data Management in Horizon 2020," 2016.

[2] E. Parliament, "Regulation (EU) 2016/679 of the European Parliament and of the Council (General Data Protection Regulation)," 2016.